EIN 4360 Practice Exam 2 **Time Limit: 50 minutes**

Instructor: Nicolas Grimaldi

QUESTIONS

Write all your answers on a separate

ANSWER SHEET

and SIGN it before starting the exam

INSTRUCTIONS

- The exam is graded out of **100 points,** yet a **maximum of 105 points** is possible.
- As promised, a **bonus 5-point question** has been tacked on at the end of the exam.
- This exam is closed book, but a one-sided sheet for notes is allowed.
- There are two sections in the exam. Section I has five (5) short free-response questions worth 40 points and section II consists of three (3) long free-response problems worth 60 points.
- Please show all your work clearly. No points will be awarded for anything that is not written in the answer.
- Sharing of calculators is not permitted.
- Please switch off your cell-phone and laptops for the duration of the exam.
- Good luck!

Section I: [40 Points] Short Free-Response Questions.

There are five (5) questions in this section each worth **8 points**. Circle the letter corresponding to the **BEST** answer on the separate ANSWER SHEET.

(Recommended) Time Limit: **20 minutes**

1. (Lecture 7) A material handling planning chart is shown below. Identify which columns relate to considerations in the material handling systems equation are shown and which of the three M's do each represent.

MATERIAL HANDLING PLANNING CHART

Company A.R.C. Inc, Prepared by I.A. Layout Alternative 1

Product Air Speed Control Value Date Sheet 1 of

Step No.	O	T	S	I	Description	Oper No.	Dept.	Cont. Type	Size	Wt.	Qty. Per Cont.	Freq	Dist	Method of Handling
1			X		Bar stock in storage (2200)		Stores.							
2		X			Profit stores to saw dept.			LDDSE (FK.TRK)	2.5″ × 3.5 × 16″	5 lb	to bars	3 times daily	16 ft	Fork lift
3			X		Store in saw department		Saw							
4	X				Cut to length	0101	Saw							
5		X			From saw to grinding			TOTE pan	15″ × 12″ × 7″	30 lb	30	Twice daily	10 ft	Platform hand truck
6			X		Store in grinding		Grinding							
7	X				Grind to length	0201	Grinding							
8			X					TOTE pan	15″ × 12″ × 7″	30 lb	30	Twice daily	13 ft	Platform hand truck
9				X	Store in deburring		Deburring							
10	X				Deburr	0301	Deburring							
11		X			From deburring to dr. Prs			TOTE pan	15″ × 12″ × 7″	30 lb	30	Twice daily	16 ft	Platform hand truck
12			X		Store in drill press		Drill Press							
13	X				Dr. CD holes tap. rean, dsk	0401	Drill Press							
14		X			From dr. press to tur. lathe			TOTE pan	15″ × 12″ × 7″	30 lb	30	Twice daily	33 ft	Platform hand truck

2. (Lecture 9) Explain why Chebyshev's measurements are considered in the calculations of the cycle times. How does this consideration impact the calculation of E[SC] and E[DC]?

3. (Lecture 6) What does the A matrix below tell us about the relative shopping enjoyment at Trader Joes (TJ), Walmart (W), and Publix (P)? Write your answer and in your (written) answer compare them (i.e., TJ vs. W, W vs. P, TJ vs. P).

A	*TJ*	*W*	*P*
TJ	1	3	1/2
W	1/3	1	1/5
P	2	5	1

4. (Lecture 5) You are assigned to design a new facility and are told that you should account for wheelchair users. The maximum expected length and width of a wheelchair are 5'6" and 2'6", respectively. Provide a justification as to what you believe the minimum aisle width should be.

5. (Lecture 8) Explain the difference between crossdocking and traditional operations and why might we use one over the other. Give an example for each.

Section II: [60 Points] Descriptive Problems.

There are three (3) main problems in this section each worth **20 points.**

(Recommended) Time Limit: **30 minutes**

Problem 1 [**20 Points**]
(Lecture 5) A particular student organization that prepares free dinners for college students every Tuesday evening has recently grown significantly in size and has asked you, the reader, to prepare a parking lot to accommodate the increased number of students. In exchange for completing this request, they will invite you to participate in those free dinners each Tuesday.

a) Determine the module type and module widths given the following information: SWc = 8', SWsd = 8.5', θ = 90°. No walls, only leftover space allocated to small walkways.
b) Find the number of each type of module that you can fit in the lot given the following information: Lot depth = 550'. Only four rows need to be compact.
c) Find the number of stalls in each row for both compact and standard cars given the following information: Lot width = 400'. Don't forget about the driving (circulation) width.
d) Find the number of ADA stalls given the following information: Because of the large size of the lot, dedicate exactly one row that would otherwise be given to standard cars as ADA instead.
e) Find the total number of parking spaces in this lot given all prior information.

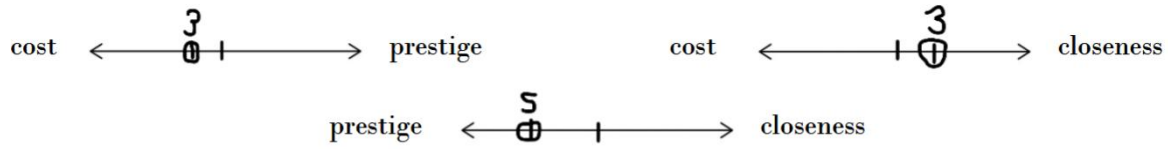
Problem 2 [**20 Points**]
(Lecture 6) Darth Vader is having a very difficult time deciding which university to send his twin children to. As a single dad, he'll be lonely without them at home, but he doesn't want his bias of having them nearby to distract him from the objectively best answer. His good friend Ben recommended he use the analytical hierarchy process to decide where to send them. Luke and Leia got as far as making the normalized comparison matrices as seen below.

cost	UF	FIU	UM
UF	0.47	0.47	0.41
FIU	0.47	0.47	0.53
UM	0.06	0.06	0.06

prestige	UF	FIU	UM
UF	0.67	0.54	0.71
FIU	0.10	0.08	0.05
UM	0.23	0.38	0.24

close to home	UF	FIU	UM
UF	0.11	0.13	0.08
FIU	0.56	0.65	0.69
UM	0.33	0.22	0.23

Here are the priority piecewise selections they all came up with as a family.



a) Assemble the priority matrix.
b) Assemble the score matrix.
c) Find the global score vector and identify which school Vader's kids should go to.

Problem 3 [20 Points]

(Lecture 9) Wachovia Bank (rest in peace) used to have a secret storage facility for customers' safety deposit boxes (SDB). As part of the financial operations team, you're collecting data on the cycle times for storing and retrieving the SDBs via storage and retrieval systems. The rack system has dimensions 20"m x 10"n, where m is 150 and n is 100. The S/R requires 0.5 min to perform a P/D operation, travels horizontally at an average speed of 11 fpm, and travels vertically at an average speed of 6 fpm.

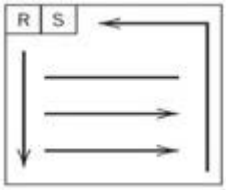
a) What are the expected travel times for single and dual command cycles?

b) What are the expected single and dual command cycle times?

Section III: [5 Points] Bonus Problem.

There is one (1) problem in this section worth **5 points.**

Bonus Problem [5 points]
(Lecture 8) What is one advantage and one disadvantage of placing receiving and storage next to each other? Note the layout below as reference.



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